1 Eigenvectors and Eigenvalues

1.1 Minimal Polynomial

1.1.1 Exercise

Exercise 1.1.1.1. Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$, and $v \in V$. Prove that:

$$\operatorname{span}(v, Tv, ..., T^m v) = \operatorname{span}\big(v, Tv, ..., T^{\dim V - 1} v\big)$$

for all integers $m \geq \dim V - 1$

Solution,

Since, $\dim V = m$, then everything after $\{v, ..., T^{\dim V - 1} v\}$ must be linearly dependent. Otherwise, it would contradict the fact that:

$$|\text{linearly dependent set}| \leq |\text{spanning set}|$$

Hence,

$$\operatorname{span}(v, Tv, ..., T^m v) = \operatorname{span}\big(v, ..., T^{\dim V - 1} v\big)$$

□

Exercise 1.1.1.2. Suppose V is a two-dimensional vector space, $T \in \mathcal{L}(V)$, and the matrix of T with respect to some basis of V is:

$$\begin{pmatrix} a & c \\ b & d \end{pmatrix}$$

(a) Show that $T^2 - (a+b)T + (ad - bc)I = 0$.

(b) Show that the minimal polynomial of T equals

$$p(z) = \begin{cases} z - a & b = c = 0 \text{ and } a = d \\ z^2 - (a+d)z + (ad - bc) & \text{otherwise} \end{cases}$$

Solution,

Given, $\dim V = 2$, and there exits a basis $\mathcal{B}$ such that:

$$[T]_{\mathcal{B}} = \begin{pmatrix} a & c \\ b & d \end{pmatrix}$$

Therefore, we first compute:

$$\begin{aligned}[T]_{\mathcal{B}}^2 - (a+b)[T]_{\mathcal{B}} + (ad-bc)I &= \begin{pmatrix} a^2+bc & ac+cd \\ ab+bd & bc+d^2 \end{pmatrix} \\ &\quad - \begin{pmatrix} a^2+ad & ac+dc \\ ab+bd & ad+d^2 \end{pmatrix} + \begin{pmatrix} ad-bc & 0 \\ 0 & ad-bc \end{pmatrix} \\ &= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}\end{aligned}$$

Therefore, it must follow:

$$T^2 - (a+b)T + (ad-bc)I = 0$$

□

Exercise 1.1.1.3. Define $T \in \mathcal{L}(\mathbb{F}^n)$ by $T(x_1, ..., x_n) = (ix_i)_i$. Find the minimal polynomial of T.

Solution,

Firstly, T has n distinct eigenvalues $\{1, ..., n\}$. Because, T is a linear operator over $\mathbb{F}^n$ this implies we have exhausted all eigenvalues of T. Thus, $\deg p = n$ where, p is the minimal polynomial. Hence,

$$p(z) = \prod_{i=1}^{n} (z - i)$$

□

Exercise 1.1.1.4. Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$ has minimal polynomial $4 + 5z - 6z^2 - 7z^3 + 2z^4 + z^5$. Find the minimal polynomial of T^{-1}.

Solution,

Here,

$$p(z) = 4 + 5z - 6z^2 - 7z^3 + 2z^4 + z^5$$

Given that the constant term of p is $4 \neq 0$, it follows that T is invertible.

By definition of the minimal polynomial,

$$p(T) \equiv 0 \implies T^{-5}p(T) \equiv 0$$

Simplifying, we get:

$$4T^{-5} + 5T^{-4} - 6T^{-3} - 7T^{-2} + 2T^{-1} + I \equiv 0$$

Hence, the polynomial:

$$q(z) = 4z^5 + 5z^4 - 6z^3 - 7z^2 + 2z + 1$$

is an annihilating polynomial of T^{-1}.

So, q is a polynomial multiple of T^{-1}'s minimal polynomial. Suppose the minimal polynomial of T^{-1} is $r(z)$.

If,

$$\deg p = \deg q \geq \deg r \implies T^{\deg r} r\big(T^{-1}\big) \equiv 0$$

By minimality, $\deg r \geq p$. Since, $\deg z^{\deg r} r\big(\frac{1}{z}\big) = \deg r$. Hence, $\deg q = \deg r = \deg p$. Therefore, the minimal polynomial of T^{-1} has the same degree as the minimal polynomial of T.

Thus, by applying uniqueness and the degree constraint, we get:

$$r(z) = z^5 + \frac{5}{4}z^4 - \frac{3}{2}z^3 - \frac{7}{4}z^2 + \frac{1}{2}z + \frac{1}{4}$$

is the minimal polynomial of T^{-1}. □

Exercise 1.1.1.5. Suppose V is finite-dimensional complex vector space with $\dim V > 0$, and $T \in \mathcal{L}(V)$. Define $f : \mathbb{C} \to \mathbb{R}$ by:

$$f(\lambda) = \dim \operatorname{range}(T - \lambda I)$$

Proof,

Now, suppose that f is continuous. Since, V is a non-trivial complex vector space it follows that T has at least one eigenvalue λ. By continuity, it would follow that there exists a ball $B_\delta(\lambda)$ such that:

$$\lambda' \in B_\delta(\lambda) \implies f(\lambda') \in B_{\frac{1}{2}}(f(\lambda))$$

However, since $\dim(\cdot) \in \mathbb{N}$, it follows that $f(\lambda') = f(\lambda)$.

By rank-nullity, this would imply that $T - \lambda' I$ is non-injective since $\dim \text{range}(T - \lambda' I) \neq \dim V$. So, it follows λ' is an eigenvalue of T.

Therefore, T has infinitely many eigenvalues concentrated in $B_\delta(\lambda)$. This contradicts finite-dimensionality of V because $\#\text{eigenvalues}(T) \leq \dim V$ where, $T \in \mathcal{L}(V)$.

So, it follows that f can't be continuous. □

Exercise 1.1.1.6. Suppose $a_0, ..., a_{n-1} \in \mathbb{F}$. Let $T \in \mathcal{L}(\mathbb{F}^n)$ whose matrix (with respect to the standard basis) is:

$$\begin{pmatrix} 0 & 0 & 0 & ... & 0 & -a_0 \\ 1 & 0 & 0 & ... & 0 & -a_1 \\ 0 & 1 & 0 & ... & 0 & -a_2 \\ \vdots & & & & & \\ 0 & 0 & 0 & ... & 0 & -a_{n-2} \\ 0 & 0 & 0 & ... & 1 & -a_{n-1} \end{pmatrix}$$

Here, all entries of the matrix are 0 except for all $1's$ on the line under the diagonal and the entries in the last column (some of which might also be zero). Show that the minimal polynomial of T is the polynomial:

$$a_0 + a_1 z + ... + a_{n-1} z^{n-1} + z^n$$

Proof,

From the matrix of T, it follows that:

$$Te_1 = e_2 \quad Te_2 = e_3 \quad ... \quad Te_{n-1} = e_n \quad Te_n = -\sum_{i=0}^{n-1} a_i e_{i+1}$$

where, (e_i) is the standard basis of $\mathbb{F}^n$.

Therefore, we have the recurrence relationship:

$$T^j e_1 = e_{j+1}$$

for $j \in (1, ..., n-1)$,

$$T^n e_1 = -\sum_{i=0}^{n-1} a_i e_{i+1}$$

Define,

$$p(z) = \sum_{i=0}^{n-1} a_i z^i + z^{n+1}$$

Hence,

$$p(T)e_1 = \sum_{i=0}^{n-1} a_i T^i e_i + T^n e_1 = \sum_{i=0}^{n-1} a_i e_{i+1} - \sum_{i=0}^{n-1} a_i e_{i+1}$$

Therefore, it follows that:

$$p(T)e_1 = 0$$

Now, consider, e_j. By the recurrence relation above,

$$e_j = T^{j-1} e_1$$

where, $2 \leq j \leq n$.

Since, polynomials of linear operators commute, it follows that:

$$p(T)e_j = p(T)T^{j-1}e_1 = T^{j-1}p(T)e_1 = 0$$

Therefore, it follows that:

$$p(T) \equiv 0$$

Hence, $p(z)$ is an annihilator of T. Hence, it must be a polynomial multiple of the minimal polynomial $r(z)$. Suppose that:

$$p(z) = q(z)r(z)$$

If we assume that $\deg p > \deg r$. We run into an issue caused by the recurrence relation above.

Since, $\deg r < n$, it follows that:

$$r(z) = \sum_{i=0}^{\deg r - 1} c_i z^i + z^{\deg r}$$

The issue that occurs is the fact that if we apply $r(T)$ to e_1, we'll get:

$$\sum_{i=0}^{\deg r - 1} c_i e_{i+1} + e_{\deg r + 1} = 0$$

Since, $\left\{e_1, ..., e_{\deg r + 1}\right\}$ is a linearly independent set in V. It follows that the above expression can't be zero since the coefficient on $e_{\deg r + 1}$ is not zero.

However, this contradicts the fact that r is the minimal polynomial so it's an annihilator of T.

Thus, we reach an contradiction, and hence $\deg r \leq \deg p \implies \deg r = \deg p$.

By uniqueness, p is the minimal polynomial.

□

Exercise 1.1.1.7. Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$ and p is the minimal polynomial of T. Suppose $\lambda \in \mathbb{F}$. Show that the minimal polynomial of $T - \lambda I$ is the polynomial q defined by:

$$q(z) = p(z + \lambda)$$

Proof,

Since, p is the minimal polynomial of T. It follows that:

$$p(T) = 0$$

Now,

$$p((T - \lambda I) + \lambda I) = 0$$

Thus, it follows that $q(z) = p(z + \lambda)$ is an annihilating polynomial of $T - \lambda I$.

Now, let $q'(z)$ be the minimal polynomial of $T - \lambda I$. Then it follows the polynomial:

$$p'(T) = q'(T - \lambda I)$$

is an annihilating polynomial of T.

The both of the crucial results above follow from the fact that $p(z + a)$ is also a polynomial. The next result we shall be using is that $\deg p(z + a) = \deg p(z + a)$ and if p is a minimal polynomial, then $\deg p \leq \deg q$ for all annihilating polynomial q.

So,

$$\deg p = \deg q \geq \deg q' = \deg p' \geq \deg p$$

Hence, the degree of the minimal polynomial q' is $\deg p$. By uniqueness, it means that q is the minimal polynomial since $p(z + \lambda)$ is a monic polynomial of $\deg q'$.

□

Exercise 1.1.1.8. Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$, ,and p is the minimal polynomial of T. Suppose $\lambda \in \mathbb{F} \setminus \{0\}$. Show that the minimal polynomial of λT is the polynomial q defined by:

$$q(z) = \lambda^{\deg p} p\left(\frac{z}{\lambda}\right)$$

Proof,

Now,

$$\lambda^{\deg p} p\left(\frac{\lambda T}{\lambda}\right) = \lambda^{\deg p} p(T) = 0$$

Therefore, $q(z)$ is an annihilating polynomial of λT. Moreover,

$$\lambda^{\deg p}\left(\frac{z}{\lambda}\right)^{\deg p} = z^{\deg p}$$

Thus, q is also monic.

Suppose q' is the minimal polynomial of λT.

Therefore,

$$p'(T) = q'(\lambda T) = 0$$

Hence, p' is an annihilating polynomial of T.

Since, scaling an input doesn't change the degree of the polynomial, we have the following inequality satisfied by the polynomial.

$$\deg p = \deg q \geq \deg q' = \deg p' \geq \deg p'$$

where, we used the fact that the degree of an annihilating polynomial is at the very least the same as the minimal polynomial.

Thanks to this, we get:

$$\deg q = \deg q'$$

Since, q is monic polynomial whose degree matches that of q', by the uniqueness of the minimal polynomial, it follows that q is the minimal polynomial of λT where $\lambda \in \mathbb{F} \setminus \{0\}$

□

Exercise 1.1.1.9. Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Let $\mathcal{E}$ be the subspace of $\mathcal{L}(V)$ defined by:

$$\mathcal{E} = \{q(T) : q \in \mathcal{P}(\mathbb{F})\}$$

Prove that $\dim \mathcal{E}$ equals the degree of the minimal polynomial of T.

Proof,

The main trick of this proof is the application of the polynomial division.

So, let p be the minimal polynomial of T. Then, by the division algorithm, we can write:

$$q(T) = d(T)p(T) + r(T)$$

Because $p(T)$ is the minimal polynomial, it follows that:

$$q(T) = r(T)$$

Due to the fact that $\deg r < \deg p \implies \operatorname{span}\{I, T, ..., T^{\deg p - 1}\} = \mathcal{E}$.

Thus,

$$\dim \mathcal{E} \leq \left|\operatorname{span}\{I, ..., T^{\deg p - 1}\}\right| = \deg p$$

To get equality, we must show that $\{I, T, ..., T^{\deg p - 1}\}$ has to be a linearly independent set.

For contradiction, assume it's linearly dependent. Therefore, there exists a set of $c_i \in \mathbb{F}$ such that:

$$\sum_{i=0}^{\deg p - 1} c_i T^i \equiv 0$$

However, this implies that the following non-zero polynomial:

$$p(z) = \sum_{i=0}^{\deg p - 1} c_i z^i$$

is an annihilating polynomial. However, it's degree is at most $\deg p - 1$ which is less than the minimal polynomial. So, our assumption that p was the minimal polynomial is contradicted.

Thus, $\{I, T, ..., T^{\deg p - 1}\}$ is linearly independent. Therefore, we get:

$$\dim \mathcal{E} = \deg p$$

□

Exercise 1.1.1.10. Suppose $T \in \mathcal{L}(\mathbb{F}^4)$ is such that the eigenvalues of T are $3, 5, 8$. Prove that:

$$(T - 3I)^2(T - 5I)^2(T - 8I)^2 = 0$$

Proof

Let p be the minimal polynomial of T.

Since, $\deg p \leq \dim V = 4$, the above polynomial which has degree greater than 4 must necessarily be a multiple of p.

Therefore, it follows that the polynomial must be annihilating.

□

Exercise 1.1.1.11. Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that the minimal polynomial of T has degree at most $1 + \dim \operatorname{range} T$

Proof,

For T, recall that generally $\operatorname{range} T$ is an invariant subspace under T.

Thus, $T|_{\operatorname{range} T} \in \mathcal{L}(\operatorname{range} T)$.

Hence, we can consider the minimal polynomial of $T|_{\operatorname{range} T}$ say $q(z)$.

Here,

$$q(T)(\operatorname{range} T) = \{0\}$$

As $\operatorname{range} T$ is a subspace of V, it follows that we can find a subspace $W \subseteq V$ such that:

$$V = \operatorname{range} T \oplus W$$

Because $q(T)$ will annihilate $\operatorname{range} T$, we only need to consider how it behaves for vectors in W.

Namely, we understand that $q(T)$ might not necessarily kill vectors in W. An intuitive way to think about it is that $q(T)$ kills vectors within $\operatorname{range} T$. If we pass w into $q(T)$, it will spend one of its powers/degrees squashing w from W to $\operatorname{range} T$. So, it won't have enough degrees remaining kill w.

An easy fix is to simply multiply $q(z)$ by z. This addition of degree allows $p(z) = q(z)z$ to spend one of its degrees squashing w into $\operatorname{range} T$ before it encounters $q(T)$. Thus allowing $q(T)$ to kill it.

Formally,

$$q(T)T(v) = q(T)(Tv)$$

For all $v \in V$.

Since, $Tv \in \operatorname{range} T \implies q(T)(Tv) = 0$.

Hence, $p(z)$ is an monic annihilating polynomial of T whose degree is at most $\dim \operatorname{range} T + 1$.

Thus, the minimal polynomial must have degree less than that. Therefore, the minimal polynomial of T is at most $\dim \operatorname{range} T + 1$.

□

Exercise 1.1.1.12. Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that T is invertible if and only if $I \in \operatorname{span}\big(T, T^2, ..., T^{\dim V}\big)$

Proof,

($\implies$) Suppose T is invertible.

Because V is finite-dimensional, T has a minimal polynomial moreover we also know it can't have a non-zero constant term. Thus,

$$p(z) = z^{\deg p} + ... + c_0$$

where, $c_0 \neq 0$.

Thus, it follows that:

$$p(T) = T^{\deg p} + ... + c_0 I = 0$$

Hence,

$$I = -\frac{1}{c_0}\big(T^{\deg} + ... + c_1 T\big)$$

Thus, $I \in \operatorname{span}(T, ..., T^n)$.

($\impliedby$) So, we've that:

$$I \in \operatorname{span}\big\{T, ..., T^{\dim V}\big\}$$

Therefore,

$$I = \sum_{n=1}^{\dim V} c_n T^n \implies \sum_{n=1}^{\dim V} c_n T^n - I = 0$$

Because of the fact that an annihilating polynomial of T has a non-zero constant term it must follow that the minimal polynomial can't have a non-zero constant term.

This follows because all annihilating polynomials are polynomial multiple of the minimal polynomial. So, if we factored the annihilating polynomial it must necessarily follow we could factor out a z if the minimal polynomial had a vanishing constant term.

Now, since the minimal polynomial of T doesn't have a zero constant term, it follows that T is invertible.

□

Exercise 1.1.1.13. Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Let $n = \dim V$. Prove that if $v \in V$, then span $\left(v, Tv, ..., T^{n-1}v\right)$ is invariant under T.

Proof,

By finite-dimensionality, we can confirm the existence of the minimal polynomial:

$$T^m + c_m T^{m-1} + ... + c_0 = 0$$

Apply, T^{n-m} to both sides, remembering that $T^0 = I$, we get:

$$T^n \in \text{span}\left(v, Tv, ..., T^{n-1}v\right)$$

Therefore, it follows that:

$$w \in \text{span}\left(v, Tv, ..., T^{n-1}v\right)$$

So,

$$Tw \in \text{span}(Tv, ..., T^n v)$$

Because, $T^n v \in \text{span}\left(v, ..., T^{n-1}v\right)$. It follows,

$$Tw \in \text{span}(v, ..., T^{n-1}v)$$

Thus, $\text{span}(v, ..., T^{n-1}v)$ is invariant under T.

□

Exercise 1.1.1.14. Suppose V is a finite-dimensional complex vector space. Suppose $T \in \mathcal{L}(V)$ is such that $5, 6$ are eigenvalues of T and T has no other eigenvalues. Prove that:

$$(T - 5I)^{\dim V - 1}(T - 6I)^{\dim V - 1} = 0$$

Proof,

The degree of the polynomial:

$$p(z) = (z - 5)^{\dim V - 1}(z - 6)^{\dim V - 1}$$

is $2(\dim V - 1)$. Since T has two eigenvalues, it follows that $\dim V \geq 2$. Thus,

$$2(\dim V - 1) \geq \dim V$$

Now, the minimal polynomial of T is only a product of $(z - 5)$ and $(z - 6)$ with varying multiplicities.

There can be at most $\dim V - 1$ $(z - 5)$ and $\dim V - 1$ $(z - 6)$ in the factorization of the minimal polynomial of T.

The polynomial p has both. Thus, it must necessarily be a polynomial multiple of the minimal polynomial of T. Therefore, it follows that:

$$(T - 5I)^{\dim V - 1}(T - 6I)^{\dim V - 1} = 0$$

□

Exercise 1.1.1.15. Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$, and U is a subspace of V that is invariant under T.

1. Prove that the minimal polynomial of T is a polynomial multiple of the operator T/U
2. Prove that:

$$(\text{minimal polynomial of } T|_U) \times (\text{minimal polynomial of } T/U)$$

is a polynomial multiple of the minimal polynomial of T

Proof,

First notation, let $p_T, p_{T|_U}, p_{T/U}$ be the minimal polynomial of $T, T|_U, T/U$ respectively.

(1.) $p_{T/U} \mid p_T$

Now,

$$\begin{aligned} p_T(T/U)(v+U) &= \sum c_n (T/U)^n (v+U) \\ &= \sum c_n (T^n v + U) \\ &= \left(\sum c_n T^n v\right) + U \\ &= p_T(Tv) + U \\ &= 0 + U \\ &= U \end{aligned}$$

Hence, p_T is an annihilating polynomial of T/U. Therefore, $p_{T/U} \mid p_T$.

> *Remark*: I ignored the indexing on the sums for simplicity. For rigor, remember it runs from 0 to $\deg p_T$

(2.) $p_T \mid p_{T|_U} p_{T/U}$

So,

$$p_{T|_U} p_{T/U}(T)v = p_{T|_U}(T)\left[p_{T/U}(Tv)\right]$$

Since,

$$p_{T/U}$$

annihilates T/U, it follows that:

$$p_{T/U}(T/U) = p_{T/U}(T)/U = U$$

This result follows from computation in (1.)

Thus, $p_{T/U}(T)v + U = U \iff p_{T/U}(Tv) \in U$.

Now, for $u \in U$,

$$p(T)|_U\ (u) = p(T)(u) \quad p(T|_U)(u) = p(T)(u)$$

Therefore,

$$p_{T|_U}(T)\Big[\Big(p_{T/U}(T)v\Big)\Big] = p_{T|_U}(T|_U)\Big[\Big(p_{T/U}(T)v\Big)\Big] = 0$$

So we're done from the same reasoning as (a). □

Exercise 1.1.1.16. Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$, and U is a subspace of V that is invariant under T. Prove that the set of eigenvalues of T equals the union of the set of eigenvalues of $T|_U$ and the set of eigenvalues of T/U.

Proof,

See the previous exercise. Because of the fact that,

$$p_T \mid p_{T|_U} p_{T/U}$$

The eigenvalues in T must be present as factors in $p_{T|_U} p_{T/U}$. However, the factors of $p_{T|_U} p_{T/U}$ is the union of factors of $p_{T|_U}$ and $p_{T/U}$ which corresponds to the union of the set of the eigenvalues of $T|_U, T/U$.

Therefore,

$$E_T \subseteq E_{T|_U} \cup E_{T/U}$$

Trivially, $E_{T|_U} \subseteq E_T$ and we also know that $E_{T/U} \subseteq E_T$. Thus,

$$E_{T|_U} \cup E_{T/U} \subseteq E_T$$

Thus,

$$E_T = E_{T|_U} \cup E_{T/U}$$

□

Exercise 1.1.1.17. Suppose $\mathbb{F} = \mathbb{R}$, V is finite-dimensional, and $T \in \mathcal{L}(V)$. Prove that the minimal polynomial of $T_{\mathbb{C}}$ equal minimal polynomial of T.

Proof,

Recall that the complexification of T is:

$$T_{\mathbb{C}}(v + iu) = Tv + iTu$$

Let, $p_{T_{\mathbb{C}}}, p_T$ be the minimal polynomial of $T_{\mathbb{C}}, T$ respectively.

Before that, let's consider how complexification affects polynomials of maps.

So, (ignoring some annoying algebra)

$$p(T_{\mathbb{C}})(v + iu) = p(T)v + ip(T)u = p(T)_{\mathbb{C}}(v + iu)$$

Therefore,

$$p(T_{\mathbb{C}}) = p(T)_{\mathbb{C}}$$

This implies that:

$$p_T(T_{\mathbb{C}}) = p_T(T)_{\mathbb{C}} = 0_{\mathbb{C}} = 0$$

Similarly,

$$p_{T_{\mathbb{C}}}(T_{\mathbb{C}}) = p(T)_{\mathbb{C}} = 0 \implies p(T) = 0$$

The implication follows since for all $v \in V, p(T)_{\mathbb{C}}v = p(T)v = 0$.

The equalities imply: $p_{T_{\mathbb{C}}} \mid p_T, p_{T_{\mathbb{C}}} \mid p_T$. So,

$$p_{T_{\mathbb{C}}} = p_T$$

Because both are monic. (If it weren't monic, then there can be an issue of ± 1 screwing things up). □

Exercise 1.1.1.18. Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that the minimal polynomial of $T' \in \mathcal{L}(V')$ equals the minimal polynomial of T.

Proof,

Again, the general thing that happens here is the fact that:

$$p(T') = p(T)'$$

Now to prove this we recall:

$$(T^n)' = \left(TT^{n-1}\right)' = (T')^{n-1}T' = (T')^n$$
$$(aT + bS)' = aT' + bS'$$

Combining this, and a bit of algebra, we can show:

$$p(T') = p(T)'$$

With this, we argue similarly to the previous exercise.

□

1.2 Upper-Triangular Matrices

Definition 1.2.1 (Matrix of an Operator): Suppose $T \in \mathcal{L}(V)$. The matrix of T with respect to a basis $B = \{v_1, ..., v_n\}$ of V is the $n \times n$ matrix:

$$[T]_B^B = \left(A_{i,j}\right)$$

where,

$$Tv_k = \sum_{i=1}^{n} A_{i,k} v_i$$

If the basis is clear from context, we'll use $[T]$ to denote the matrix.

□

A central goal of linear algebra is to show that given an operator T on a finite-dimensional vector space V, there exists a basis of V with respect to which T has a reasonably simple matrix.

□

Definition 1.2.2 (Diagonal of a Matrix): The diagonal of a square matrix consists of the entries on the line from the upper left corner to the bottom right corner.

So if $A = \left(A_{i,j}\right)$, then the diagonal of A are the elements of the form $A_{i,i}$

□

Definition 1.2.3 (Upper Triangular Matrix): A square matrix is called upper triangular if all entries below the diagonal are 0.

□

Theorem 1.2.1 (Conditions for Upper-Triangular Matrix): Suppose $T \in \mathcal{L}(V)$ and $B = (v_1, ..., v_n)$ be a basis of V. Then the following are equivalent:

1. $[T]_B^B$ is upper triangular
2. $\text{span}(v_1, ..., v_k)$ is invariant under T for $k \in \{1, ..., n\}$
3. $Tv_k \in \text{span}(v_1, ..., v_k)$ for each $k = 1, ..., n$

Proof,

$(1. \implies 2.)$

Here, $[T]_B^B$ is upper triangular. Therefore,

$$[T]_B^B = (A_{i,j}) \implies A_{i,j} = \begin{cases} 0 & i > j \\ A_{i,j} & i \leq j \end{cases}$$

Therefore,

$$Tv_l = \sum_{i=1}^{l} A_{i,k} v_i$$

Hence, for $l \leq k$,

$$Tv_l \in \text{span}(v_1, ..., v_k)$$

Thus, $\text{span}(v_1, ..., v_k)$ is invariant under T for $k = 1, ..., n$.

$(2. \implies 3.)$.

Since, $\text{span}(v_1, ..., v_k)$ is invariant under T for $k \in \{1, ..., n\}$. It follows that,

$$Tv_k \in \text{span}(v_1, ..., v_k)$$

Finally, $(3. \implies 1.)$

Because,

$$Tv_k = \sum_{i=1}^{k} A_{i,k} v_k$$

We've, if $i > j$,

$$A_{i,j} = 0$$

So, $[T]_B^B$ satisfies the definition of being upper-triangular.

□

Theorem 1.2.2 (Equation satisfied by operator with upper-triangular matrix): Suppose $T \in \mathcal{L}(V)$ and V has a basis with respect to which T has an upper-triangular matrix with diagonal entries $\lambda_1, ..., \lambda_n$. Then,

$$\prod_{i=1}^{n} (T - \lambda_i I) = 0$$

Proof,

Let $v_1, ..., v_n$ denote a basis of V with respect to which $[T]$ is upper-triangular with diagonal entries $\lambda_1, ..., \lambda_n$.

Then,

$$Tv_1 = \lambda_1 v_1$$

which means that $(T - \lambda_1 I)v_1 = 0$, which implies that:

$$\prod_{i=1}^{m}(T-\lambda_i I)v_1 = 0$$

For $m = 1, ..., n$. (Here, we used the fact that polynomial of operators commute).

Now,

$$Tv_2 = \lambda_2 v_2 + cv_1 \implies (T-\lambda_2 I)v_2 \in \text{span}(v_1)$$

This follows from the previous theorem.

Therefore,

$$(T-\lambda_1 I)(T-\lambda_2 I)v_2 = 0$$

Since, $(T-\lambda_1 I)(v)$ annihilates every vector in $\text{span}(v_1)$.

Now, we continue like this. Suppose that for $k \geq 2$,

$$\prod_{i=1}^{k}(T-\lambda_i I)(\text{span}(v_1, ..., v_k)) = \{0\}$$

So,

$$(T-\lambda_{k+1} I)v_{k+1} \in \text{span}(v_1, ..., v_k)$$

By our inductive hypothesis, we get:

$$\prod_{i=1}^{k+1}(T-\lambda_i I)(\text{span}(v_1, ..., v_k)) = \{0\}$$

Thus, by induction, we've:

$$\prod_{i=1}^{n}(T-\lambda_i I)(\text{span}(v_1, ..., v_n)) = \{0\}$$

As, $V = \text{span}(v_1, ..., v_n)$ the result follows.

□

Theorem 1.2.3 (Determination of eigenvalues from upper-triangular matrix): Suppose $T \in \mathcal{L}(V)$ has an upper-triangular matrix with respect to some basis of V. Then the eigenvalues of T are precisely the entries on the diagonal of the upper-triangular matrix.

Proof,

Suppose $v_1, ..., v_n$ is a basis of V with respect to which $[T]$ is an upper triangular matrix with diagonals $\lambda_1, ..., \lambda_n$.

Because, $Tv_1 = \lambda_1 v_1$, we see that λ_1 is an eigenvalue of T.

Suppose $k \in \{2, ..., n\}$. Then, $(T - \lambda_k I)v_k \in \text{span}(v_1, ..., v_{k-1})$. Thus,

$$(T - \lambda_k I)(\text{span}(v_1, ..., v_k)) = \text{span}(v_1, ..., v_{k-1})$$

Because,

$$\dim \text{span}(v_1, ..., v_k) = k \quad \dim \text{span}(v_1, ..., v_{k-1}) = k - 1$$

By rank-nullity, it follows that $T - \lambda_k I$ is non-injective. Thus, λ_k is an eigenvalue of T.

Therefore, we've shown that every diagonal element of $[T]$ is an eigenvalue of T.

Now, we have to show that there are no other eigenvalues of T.

Let q be the polynomial defined by:

$$q(z) = \prod_{i=1}^{n} (z - \lambda_i)$$

By the previous theorem,

$$q(T) \equiv 0$$

Thus, q is a polynomial multiple of the minimal polynomial. The minimal polynomial contains within its factors all eigenvalues of T. Thus, all eigenvalues must be within $\{\lambda_1, ..., \lambda_n\}$.

Thus, we're done.

□

Theorem 1.2.4 (Necessary and sufficient condition to have an upper-triangular matrix): Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Then T has an upper-triangular matrix with respect to some basis of V if and only if the minimal polynomial of T equals:

$$p(z) = \prod_{i=1}^{m}(z - \lambda_i)$$

for some $\lambda_1, ..., \lambda_m \in \mathbb{F}$

Proof,

($\Longrightarrow$) Suppose that T has a basis with which $[T]$ is upper triangular. Then, suppose that $\{\lambda_i\}_{i \leq n}$ be the elements on $[T]$'s diagonal. It then follows,

$$q(T) = \prod_{i=1}^{n}(T - \lambda_i I) = 0$$

Since, $q(z)$ is polynomial-multiple of $p(z)$ the minimal polynomial of T. It follows that,

$$p(z) = \prod_{i=1}^{m}\Big(z - \lambda_{\sigma(i)}\Big)$$

where, σ is a permutation.

($\Longleftarrow$) Suppose that the minimal polynomial of T is:

$$p(z) = (z - \lambda_1)...(z - \lambda_m)$$

We shall proceed by induction on m.

If $m = 1$, then:

$$T = \lambda_1 I$$

Thus, the matrix of T with respect to any basis is going to be upper-triangular.

Now, we're going to proceed by strong induction.

Suppose $m \geq 1$ and the desired result holds for all smaller integers.

Let,

$$U = \text{range}(T - \lambda_m I)$$

Since,

$$p(z) = z - \lambda_m \implies \text{range}\, p(T)$$

is invariant under T. And,

$$\text{range}\, p(T) = \{Tv - \lambda_m v : v \in V\} = U$$

Therefore, U is invariant under T therefore,

$$T|_U \in \mathcal{L}(U)$$

If $u \in U$, then $u = (T - \lambda_m I)v$ for some $v \in V$. This is by definition.

Now,

$$\prod_{i=1}^{m-1} (T - \lambda_i I)u = \prod_{i=1}^{m} (T - \lambda_i I)v = 0$$

By our induction hypothesis, there is a basis $u_1, ..., u_M$ of U with respect to which $T|_U$ has an upper-triangular matrix. Thus, for each $k \in \{1, ..., M\}$ we have:

$$Tu_k = (T|_U)(u_k) \in \text{span}(u_1, ..., u_k)$$

Extend $u_1, ..., u_M$ to a basis $u_1, ..., u_M, v_1, ..., v_N$ of V. If $k \in \{1, ..., N\}$, then:

$$Tv_k = (T - \lambda_m I)v_k + \lambda_m v_k$$

The definition of U shows that $(T - \lambda_m I)v_k \in U$. Thus,

$$Tv_k \in \text{span}(1, ..., u_m, v_1, ..., v_k)$$

Thus, we conclude that T has an upper triangular matrix with respect to the basis $u_1, ..., u_M, u_1, ..., v_N$.

□

Corollary 1.2.4.1 *(If $\mathbb{F} = \mathbb{C}$, then every operator on V has an upper-triangular matrix)*: Suppose V is a finite-dimensional complex vector space and $T \in \mathcal{L}(V)$. Then, T has an upper triangular matrix with respect to some basis of V.

Proof,

Follows from the previous theorem and the fundamental theorem of algebra.

□
